

The Exact Minimax Two-Step Schedule for Smooth Convex Gradient Descent

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Abstract

We determine the exact worst-case optimal pair of two positive, fixed gradient-descent steps for differentiable convex functions with Lipschitz gradient. If the steps are normalized by the smoothness constant, the ordered pair

$$\left(\sqrt{2}, \frac{3 + \sqrt{9 + 8\sqrt{2}}}{4}\right)$$

attains the minimax final objective gap

$$\frac{1}{5 + 4\sqrt{2} + \sqrt{9 + 8\sqrt{2}}}.$$

The global lower bound is supplied by four explicit one-dimensional functions whose strict sublevel conditions partition the entire positive step quadrant. The matching upper bound is a dimension-free smooth-convex interpolation certificate with six positive multipliers and a rank-one positive-semidefinite residual. No uniqueness claim is made.

1 Problem and context

For $L > 0$, let $\mathcal{F}_L(\mathbb{R}^d)$ be the class of differentiable convex functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ whose gradients are L -Lipschitz and whose minimizer sets are nonempty proper subsets of $\mathbb{R}^d$. Given positive fixed steps η_1, η_2 , set

$$x_1 = x_0 - \eta_1 \nabla f(x_0), \quad x_2 = x_1 - \eta_2 \nabla f(x_1),$$

and define

$$R_2^L(\eta_1, \eta_2) = \sup_{\substack{d \geq 1, f \in \mathcal{F}_L(\mathbb{R}^d), \\ x_* \in \arg \min f, \\ x_0 \notin \arg \min f}} \frac{f(x_2) - f(x_*)}{L \|x_0 - x_*\|^2}. \quad (1)$$

The pair is ordered and chosen independently of the objective.

Nonconstant fixed steps can improve gradient descent on smooth convex objectives. The branch-and-bound performance-estimation computations of Das Gupta, Van Parys, and Ryu report, for two steps, the numerical pair (1.414214, 1.876768) and value 0.065946 [2, Tables 11–12]. Grimmer, Shu, and Wang give the corresponding radical pair in their table of basic f -composable schedules; their composability theory provides a tight fixed-schedule objective guarantee and motivates a general minimax conjecture for basic schedules [1, Definition 1 and Section 3.2.2]. Jung, Cho, and Yun

study the same dimension-free performance measure and establish general schedule lower bounds [3, Equation (1) and Theorem 3.3]. The argument below is an exact, self-contained resolution of the two-step problem: the essential additional ingredient is a lower bound valid simultaneously for every point of the unbounded positive step quadrant.

Theorem 1. *For every $L > 0$,*

$$\inf_{\eta_1, \eta_2 > 0} R_2^L(\eta_1, \eta_2) = C_* := \frac{1}{5 + 4\sqrt{2} + \sqrt{9 + 8\sqrt{2}}}.$$

The infimum is attained by the ordered normalized schedule

$$L\eta_1 = \sqrt{2}, \quad L\eta_2 = \frac{3 + \sqrt{9 + 8\sqrt{2}}}{4}.$$

No assertion of uniqueness is intended.

2 Normalization and exact constants

Put $h_i = L\eta_i$ and replace f by f/L . The resulting function is 1-smooth and the iterates are unchanged. For a selected minimizer x_* , let $D = \|x_0 - x_*\| > 0$ and define

$$z = \frac{x - x_*}{D}, \quad \phi(z) = \frac{f(x_* + Dz) - f(x_*)}{D^2}.$$

Then ϕ is convex and 1-smooth, the selected minimizer becomes the origin, the initial distance is one, and the normalized gradient steps are h_1, h_2 . Consequently it suffices to set

$$L = 1, \quad x_* = 0, \quad f_* = 0, \quad \|x_0\| = 1$$

and optimize over positive normalized steps. We write the resulting worst-case quantity as $R(a, b)$.

Set

$$s = \sqrt{2}, \quad t = \sqrt{9 + 8s}, \quad r = \frac{2}{1 + t}, \quad C = \frac{r^2}{2}, \quad (2)$$

and define

$$u_0 = s - 1, \quad v_0 = r(1 + s), \quad H = 2 + u_0 + v_0. \quad (3)$$

The relations $s^2 = 2$ and $t^2 = 9 + 8s$ give

$$C = \frac{1}{5 + 4s + t}, \quad 1 + v_0 = \frac{3 + t}{4}, \quad (4)$$

$$u_0 v_0 = r, \quad v_0 = r(u_0 + 2), \quad H = s + (1 + v_0) = \frac{r^{-2} - 1}{2}. \quad (5)$$

Moreover $0 < r < 1$.

3 Four witnesses and the global lower bound

The next lemma gives four genuine objective values, rather than relaxations of the worst-case problem.

Lemma 2 (Explicit adversaries). *For every $a, b > 0$,*

$$R(a, b) \geq \max \left\{ \frac{1}{2(1+2(a+b))}, \quad \frac{((a-1)(b-1))^2}{2}, \quad \frac{(b-1)^2}{2(a+1)^2} \right\}. \quad (6)$$

If $a > 1$, then also

$$R(a, b) \geq \frac{(a-1)^2}{2(b+1)^2}. \quad (7)$$

Proof. All four constructions are one-dimensional, start from $x_0 = 1$, and have the unique minimizer 0.

First take $\delta = (1 + 2(a + b))^{-1}$ and the two-sided Huber function

$$H_\delta(x) = \begin{cases} x^2/2, & |x| \leq \delta, \\ \delta|x| - \delta^2/2, & |x| \geq \delta. \end{cases}$$

The first two iterates satisfy

$$x_1 - \delta = \frac{a + 2b}{1 + 2(a + b)} > 0, \quad x_2 - \delta = \frac{a + b}{1 + 2(a + b)} > 0.$$

Thus both gradients used by the method equal δ , and direct evaluation gives $H_\delta(x_2) = 1/(2(1 + 2(a + b)))$.

For $f(x) = x^2/2$, one has $x_2 = (1 - a)(1 - b)$, giving the second term in (6). For the third, let $\delta = (1 + a)^{-1}$ and use

$$P_\delta^+(x) = \begin{cases} x^2/2, & x \leq \delta, \\ \delta x - \delta^2/2, & x \geq \delta. \end{cases}$$

Since $(P_\delta^+)'(x) = \min\{x, \delta\}$, the iterates are $x_1 = \delta$ and $x_2 = \delta(1 - b) \leq \delta$, so $P_\delta^+(x_2) = (b - 1)^2/(2(a + 1)^2)$.

Finally suppose $a > 1$, set $\delta = (a - 1)/(b + 1)$, and take

$$P_\delta^-(x) = \begin{cases} -\delta x - \delta^2/2, & x \leq -\delta, \\ x^2/2, & x \geq -\delta. \end{cases}$$

Here $(P_\delta^-)'(x) = \max\{x, -\delta\}$. Hence $x_1 = 1 - a = -(b + 1)\delta < -\delta$ and $x_2 = -\delta$, proving (7). Each displayed derivative is continuous, nondecreasing, and 1-Lipschitz. The functions therefore belong to the normalized admissible class. $\square$

Proposition 3 (Positive-quadrant partition). *For every ordered pair $a, b > 0$, one has $R(a, b) \geq C$.*

Proof. Assume for contradiction that $R(a, b) < C$, and put

$$u = a - 1 > -1, \quad v = b - 1 > -1.$$

The Huber, quadratic, and right-capped bounds in Lemma 2, together with (5), imply

$$a + b > H, \quad |uv| < r, \quad |v| < r(u + 2). \quad (8)$$

When $u > 0$, the left-capped bound also gives

$$u < r(v + 2). \quad (9)$$

Suppose first that u, v are not both nonnegative. If $u < 0 \leq v$, then (8) gives $v < r(u + 2) < 2r$ and hence $u + v < 2r$. If $v < 0 < u$, equation (9) similarly gives $u < r(v + 2) < 2r$ and $u + v < 2r$; the case $u = 0 > v$ is immediate. If both are negative, the same conclusion is immediate. But

$$H - (2 + 2r) = u_0 + v_0 - 2r = (s - 1)(1 + r) > 0,$$

so $a + b = 2 + u + v < 2 + 2r < H$, contradicting (8).

It remains to consider $u, v \geq 0$. Suppose first that $v \geq u$. If $u \leq u_0$, then

$$u + v < (1 + r)u + 2r \leq (1 + r)u_0 + 2r = u_0 + v_0. \quad (10)$$

If $u > u_0$, then $uv < r$ and $v \geq u$ imply $u < \sqrt{r}$ and

$$u + v < u + \frac{r}{u}.$$

The map $z \mapsto z + r/z$ is strictly decreasing on $(0, \sqrt{r})$. Also $u_0 < \sqrt{r}$: indeed, $u_0^2 = 3 - 2s < 1/5$, while $t < 5$ gives $r > 1/3$. It follows that

$$u + v < u_0 + \frac{r}{u_0} = u_0 + v_0. \quad (11)$$

When $u \geq v$, the same two cases with u and v interchanged use (9) in place of the right-capped inequality; if $u = v = 0$ the conclusion is immediate. Thus in every case $u + v < u_0 + v_0$, whence $a + b < H$, again contradicting (8). The assumed strict sublevel set is empty, proving the proposition. $\square$

4 A dimension-free upper certificate

We use the following elementary smooth-convex interpolation inequality.

Lemma 4. *If f is convex and 1-smooth, $f_i = f(x_i)$, and $g_i = \nabla f(x_i)$, then for every ordered pair of sample points,*

$$I_{ij} := f_i - f_j - \langle g_j, x_i - x_j \rangle - \frac{1}{2} \|g_i - g_j\|^2 \geq 0. \quad (12)$$

Proof. Let $q(x) = f(x) - \langle g_j, x \rangle$. This function is convex and 1-smooth, and $\nabla q(x_j) = 0$, so x_j minimizes q . The descent lemma at x_i yields

$$q(x_i - \nabla q(x_i)) \leq q(x_i) - \frac{1}{2} \|\nabla q(x_i)\|^2.$$

The left side is at least $q(x_j)$, which is exactly (12) after rearrangement. $\square$

Now take the ordered schedule

$$a = s, \quad b = 1 + v_0 = \frac{3 + t}{4}. \quad (13)$$

Use sample indices $\{*, 0, 1, 2\}$, with $x_* = g_* = 0$, $f_* = 0$, and

$$x_1 = x_0 - sg_0, \quad x_2 = x_1 - (1 + v_0)g_1.$$

Assign the following multipliers to six directed interpolation inequalities; all omitted multipliers are zero.

(i, j)	λ_{ij}	(i, j)	λ_{ij}	(i, j)	λ_{ij}
$(*, 0)$	sr^2	$(*, 1)$	$(2 + s)r^2$	$(*, 2)$	r
$(0, 1)$	$(1 + s)r^2$	$(1, 0)$	r^2	$(1, 2)$	$1 - r$

They are positive because $0 < r < 1$. Direct expansion gives the exact identity

$$C \|x_0\|^2 - f_2 = \sum_{i,j} \lambda_{ij} I_{ij} + \left\| \frac{r}{s} x_0 - r g_0 - v_0 g_1 - \frac{1}{s} g_2 \right\|^2. \quad (14)$$

For completeness, the finite coefficient check behind (14) can be read as follows. In the ordered Gram basis $z = (x_0, g_0, g_1, g_2)$, expansion of the interpolation sum has function-value coefficient vector $(0, 0, -1)$ for (f_0, f_1, f_2) . If M is its symmetric Gram coefficient matrix, $e_0 = (1, 0, 0, 0)^\top$, and

$$w = \left(\frac{r}{s}, -r, -v_0, -\frac{1}{s} \right)^\top,$$

then substitution of (5) gives

$$C e_0 e_0^\top - M = w w^\top = \begin{pmatrix} r^2/2 & -r^2/s & -r v_0/s & -r/2 \\ -r^2/s & r^2 & r v_0 & r/s \\ -r v_0/s & r v_0 & v_0^2 & v_0/s \\ -r/2 & r/s & v_0/s & 1/2 \end{pmatrix}. \quad (15)$$

Equations (15) and the function-value balance are precisely (14), coefficient by coefficient.

Every term on the right of (14) is nonnegative by Lemma 4. Therefore

$$f(x_2) - f_* = f_2 \leq C \|x_0 - x_*\|^2 \quad (16)$$

for every finite dimension at the schedule (13).

5 Conclusion

Proposition 3 proves that every positive ordered schedule has worst-case value at least C . At the schedule (13), inequality (16) gives the reverse bound. In fact, each of the Huber, quadratic, and right-capped witnesses is tight there, by (5). Undoing the normalization and using (4) proves Theorem 1.

The result concerns exactly two positive, deterministic, nonadaptive gradient steps and the final objective gap in (1). It does not establish uniqueness of the minimizing pair, nor does it address longer schedules, adaptive rules, or methods with momentum.

Computational and AI assistance. A numerical four-sample performance-estimation calculation was used only to identify a candidate support for the six multipliers in (14). Exact radical simplification and every Gram-matrix coefficient in (15) were then checked symbolically with SymPy. A large language model assisted with algebraic exploration, proof organization, and manuscript preparation. The computations are not used as substitutes for proof: the complete quadrant argument and the exact certificate are displayed above. The author reviewed the argument and assumes responsibility for the manuscript.

References

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